PERSONAL REVIEW NOTES

Tabular & Time-Series Data

A structured summary of data types, common data errors, and how each maps to a machine learning problem — with diagrams for every concept.

Based on the AI VIETNAM course "Tabular & Time-series Data" (M01W01)

Compiled for personal review — August 2026

Contents

Part I

Introduction to Data Science

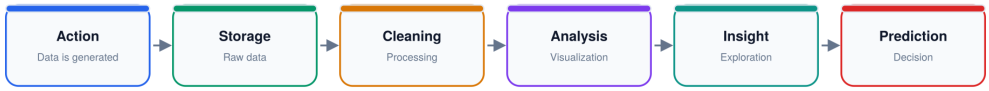
1. Data Science definition

Data Science is the process of using data to understand a real-world problem, uncover patterns, generate insight, and support decisions. Data is generated continuously in the Big Data era — from IoT devices (sensors, cameras), e-commerce platforms (Shopee, Lazada), banking transactions, and learning platforms.

The goal of Data Science is to turn raw **Data** into useful **Information**, and from there into real business **Action** — decisions or strategy.

The process runs through six stages: **Collect → Store → Clean → Analyze → Explore → Predict**

- **Collection & Storage** — capture data from real actions and organize it so it can be queried later.
- **Cleaning** — standardize the data and handle errors or missing values before it reaches a model.
- **Analysis & Exploration** — use charts to see the big picture and understand the data's characteristics.
- **Prediction** — apply a machine learning or deep learning model to forecast future trends and support decisions.



Real-world example

Raw data (Input)	Analysis / Processing (Data Science)	Result (Action)
12 months of order history	Frequency & seasonal best-sellers analysis	Optimized restocking plan
Browsing & click history	Train an ML model to learn preferences	Personalized recommendations
Bank card transactions	Build an anomaly-detection algorithm	Block fraudulent transactions

Fig. 1 — The Data Science pipeline: from action to prediction, with a real-world example.

2. Four levels of analysis

Four levels of analysis, viewed across time — Past, Present, Future:

- **Descriptive** (what happened) & **Diagnostic** (why it happened) — look at the past/present. This is classic **Business Intelligence**.
- **Predictive** (what will happen) & **Prescriptive** (what we should do) — look toward the future. This is where **Machine Learning and AI** take over.

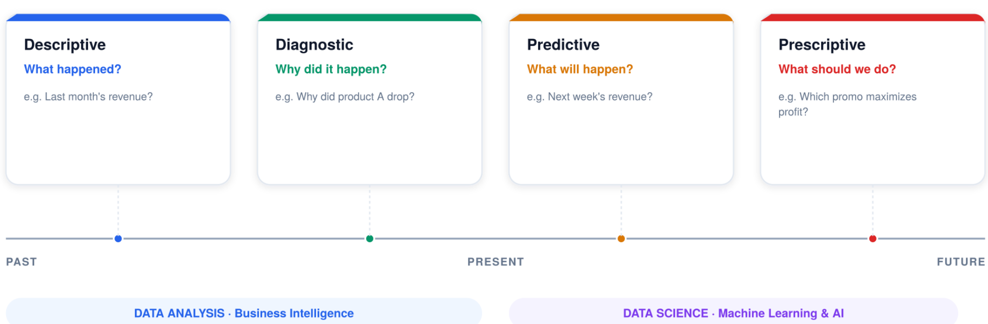


Fig. 2 — The four levels of analytics placed on a past-to-future timeline.

KEY WORDS

descriptive · diagnostic · predictive · prescriptive · business intelligence · Machine Learning and AI

3. Data types in Data Science

Four common data types (you can also explore Voice or Graph data on your own if interested):

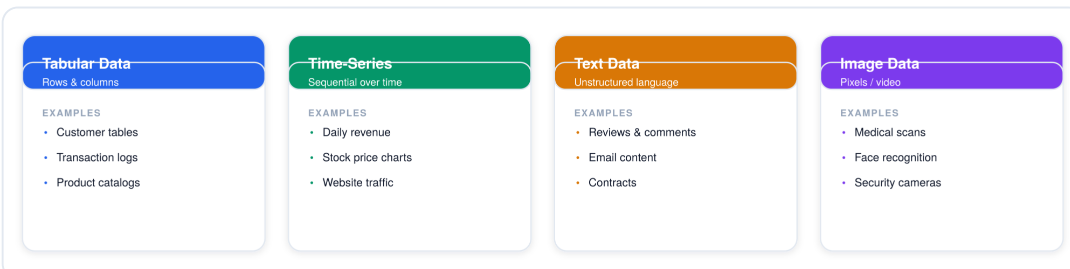


Fig. 3 — Four common data types: Tabular, Time-Series, Text, and Image data.

KEY WORDS

tabular data · time-series · text data · image data

4. Data in AI

For an AI model to predict well, it first needs data that is relevant, sufficient in quality, and correctly processed.

Data in AI: Input data → Feature + Label → used to train a predictive model.

It's like telling the machine: "OK, based on the patterns and rules I'm giving you to learn, now try to predict this new case for me" — that simple.

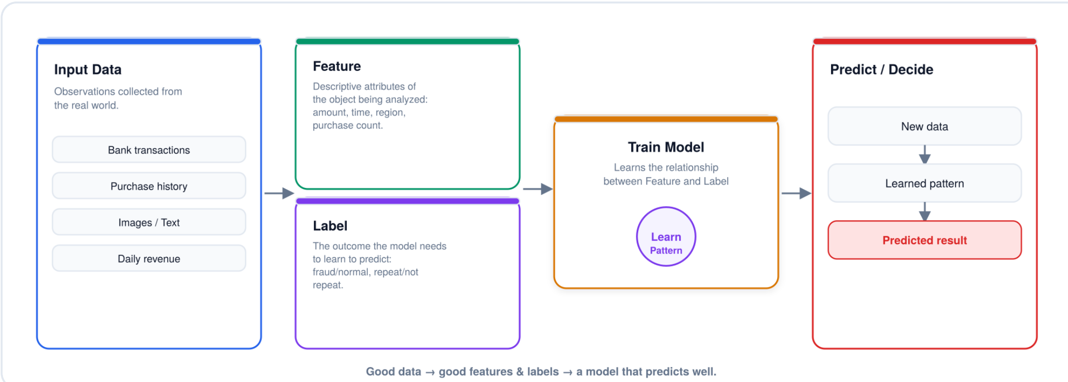


Fig. 4 — Data-in-AI pipeline: input data splits into Feature and Label, trains a model, and produces a prediction.

KEY WORDS

input data · feature · training model · prediction / decision

Part II

Tabular Data

1. What is Tabular Data?

Tabular data structure (Row, Column, Cell) is usually stored as a CSV file, Excel, a SQL database, Parquet, or a data warehouse.

- **Row / Record** — a single record, usually representing one object or one event. *e.g. one customer, one order, one transaction.*
- **Column / Dimension** — a field describing one attribute of the data. *e.g. age, city, order count, total spend.*
- **Cell / Value** — a specific value at the intersection of a row and column. *e.g. customer C003 has Total_Spent = 2,800,000.*

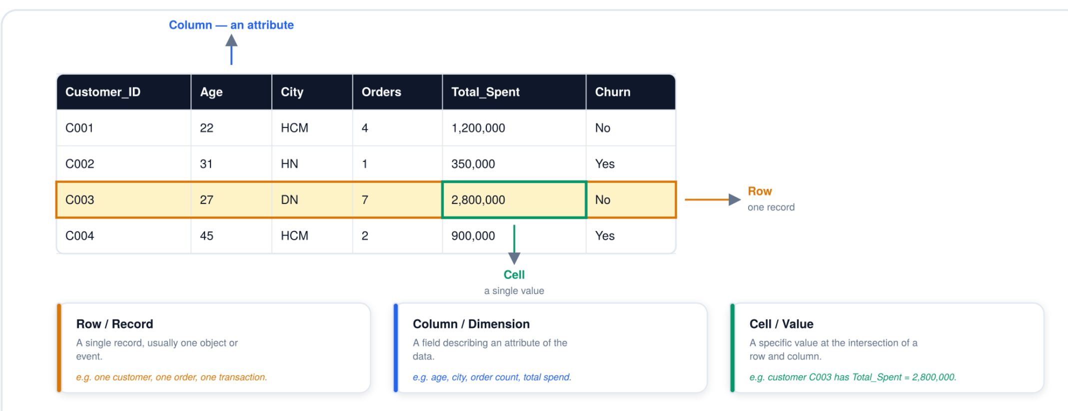


Fig. 5 — Tabular data structure showing Row, Column, and Cell with real customer data.

KEY WORDS

tabular data definition · row/record · column/dimension · cell/value · Data format: CSV, XLSX, Parquet · Processing tool: Excel, Google Sheets, Pandas · Storage system: SQL, Data Warehouse

2. Granularity (level of detail)

Granularity refers to how deep the data is observed. We can view it broadly at the total-customer level, or more specifically down to each order, or each product inside that order.

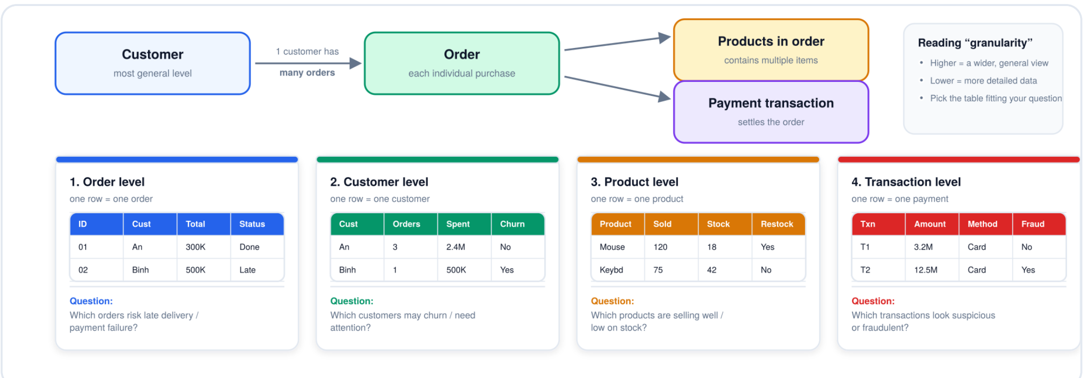


Fig. 6 — Granularity hierarchy from Customer down to Order, Products, and Payment, with four example tables at different levels of detail.

3. Data types in Tabular Data

Component data types inside a tabular dataset: **Numerical, Categorical, Boolean, Datetime.**

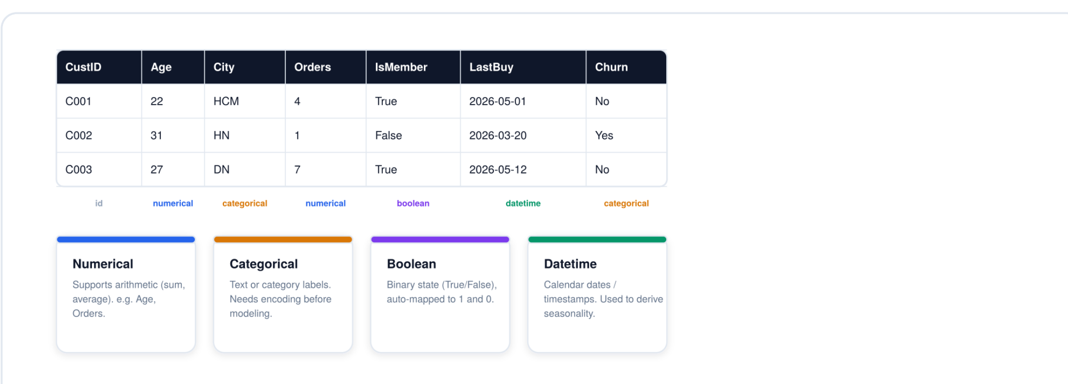


Fig. 7 — A mixed-type customer table with columns tagged as numerical, categorical, boolean, and datetime.

KEY WORDS

numerical · categorical · boolean · datetime

4. Common errors in tabular data

Error cases and how to handle them:

- **Missing values** — some cells have no value at all.
- **Duplicate cases** — one or more rows are repeated. There are several distinct duplicate scenarios, each needing its own fix.
- **Outlier** — a value that sits far outside the rest of the data. Needs to be identified and handled (data-entry error vs. a valid spike).

CustID	Age	City	Orders	Spent	LastBuy	IsMember	Churn
C001	22	HCM	4	1.2M	3 days	True	No
C002	(missing)	HN	1	350K	45 days	False	Yes
C003	27	(missing)	7	2.8M	2 days	True	No
C004	45	HCM	2	900K	60 days	False	Yes
C004	45	HCM	2	900K	60 days	False	Yes
C005	250	CT	5	99M	5 days	True	No

Missing value — an empty cell

Duplicate — a repeated row

Outlier — an unusually extreme value

Fig. 8 — A customer table highlighting missing values, a duplicated row, and an outlier.

4a. Dealing with missing values

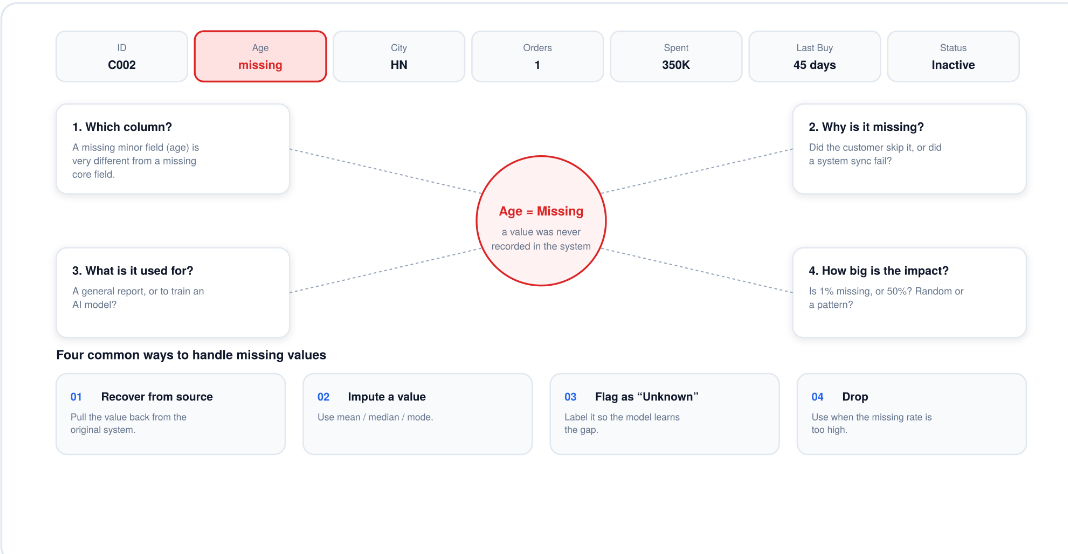


Fig. 9 — Diagnostic questions for a missing value, plus four common ways to handle it.

#	Approach	When to use it
1	Recover from source	Pull the value back from the original system.
2	Impute a value	Replace with mean, median, or mode.
3	Flag as "Unknown"	Label it so the model can learn the gap itself.
4	Drop	Use when the missing rate is too high to trust.

4b. Duplicate cases and solutions

a. **Re-imported data** — a row appears twice because a file was imported or crawled twice. **Fix: keep one representative row, delete the exact duplicate.**

b. **Same order, different status** — the system stores multiple records as an order gets updated over time. **Fix: keep the latest version, or check the update history.**

c. **Duplicate customer profile** — two profiles created from different name spellings but the same phone/email. **Fix: merge profiles, or pick one canonical record.**

d. **Near-duplicate, different OrderID** — two rows look identical except the system generated a new order code. **Fix: check PaymentID — merge/delete if it's the same transaction.**

e. **Data-entry / system error** — a mistyped number, etc. **Fix: correct the value, or set to Null if it can't be recovered.**

f. **A real spike, still valid** — a customer bulk-buying, or a revenue surge during a Flash Sale. **Fix: keep it — it's not an error.**

g. **Mismatched unit / format** — data merged from two sources using different units. **Fix: convert everything to one consistent unit of measurement.**

Case 1 — Re-imported data

An order row appears twice because the file was imported or crawled twice.

OrderID	Customer	Product	Total
O102	An	Keyboard	750K
O102	An	Keyboard	750K

Fix: keep one representative row, delete the exact duplicate.

Case 2 — Same order, different status

The system stores multiple records as the order gets updated over time.

Version	OrderID	Status	Updated
Old	O215	Pending	09:10
New	O215	Done	09:45

Fix: keep the latest version, or check the update history.

Case 3 — Duplicate customer profile

Two profiles are created from different name spellings but the same phone/email.

CustomerID	Name	Phone
C018	Nguyen An	0901234567
C077	Nguyen An	0901234567

Fix: merge profiles, or pick one canonical record.

Case 4 — Near-duplicate, different OrderID

Two rows look identical except the system generated a new order code.

OrderID	Customer	Total	Payment
O301	Binh	300K	P888
O777	Binh	300K	P888

Fix: check PaymentID — merge/delete if it's the same transaction.

Fig. 10 — Four common duplicate-data cases, each with its own fix.

4c. Outliers

Order ID	Customer	Value (VND)
ORD-01	Retail	150,000
ORD-02	Retail	320,000
ORD-03	Retail	500,000
ORD-04	Business	99,000,000
ORD-05	Retail	200,000

1. Data-entry / system error

Extra zero typed (990K → 99M), or a technical bug (Age = 250).

Fix: correct the value, or set to Null if it can't be recovered.

2. A real spike — still valid

A B2B customer bulk-buying, or a Flash Sale revenue spike.

Fix: keep it! Split into a separate segment (e.g. VIP customers).

3. Mismatched unit / format

Data merged from two sources: one in VND, another in thousands of VND.

Fix: convert everything to one consistent unit.

Fig. 11 — An outlier order value on a number line, with three possible causes.

5. Visualization using charts

A few common analysis charts: **Bar chart** (compares between groups), **Histogram** (shows the distribution of numeric data), **Boxplot** (finds outliers). There are, of course, many other chart types too.

The same table can be used to answer very different business questions — and each question calls for a different chart:

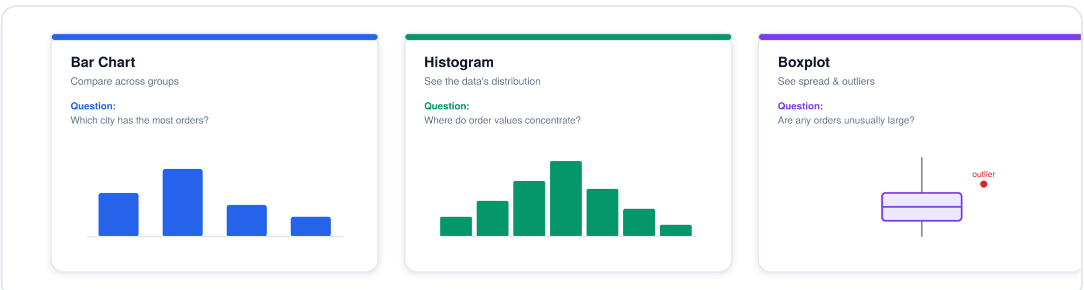
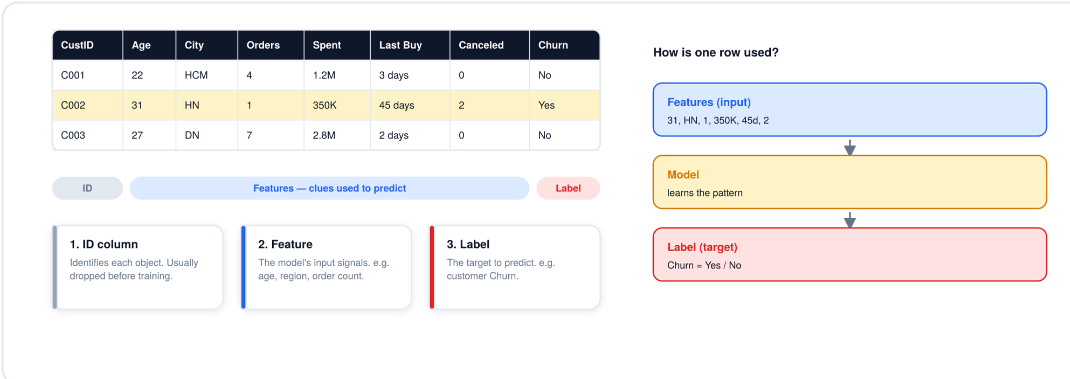


Fig. 12 — Bar Chart, Histogram, and Boxplot compared, each answering a different type of business question.

6. Feature and Label in Tabular Data

Every row used to train a model breaks down into three roles: an **ID column** (identifies each object, usually dropped before training), a **Feature** (the model's input signal), and a **Label** (the target the model needs to learn to predict).



CustID	Age	City	Orders	Spent	Last Buy	Canceled	Churn
C001	22	HCM	4	1.2M	3 days	0	No
C002	31	HN	1	350K	45 days	2	Yes
C003	27	DN	7	2.8M	2 days	0	No

Fig. 13 — A churn-prediction dataset broken into ID, Feature, and Label columns, with the row-to-prediction pipeline.

KEY WORDS

ID · Feature · Label

7. Machine Learning in Tabular Data

The same tabular structure supports different ML problems depending on the question asked of it — whether we need to classify, predict a number, or group the data.

Machine Learning has three broad branches:

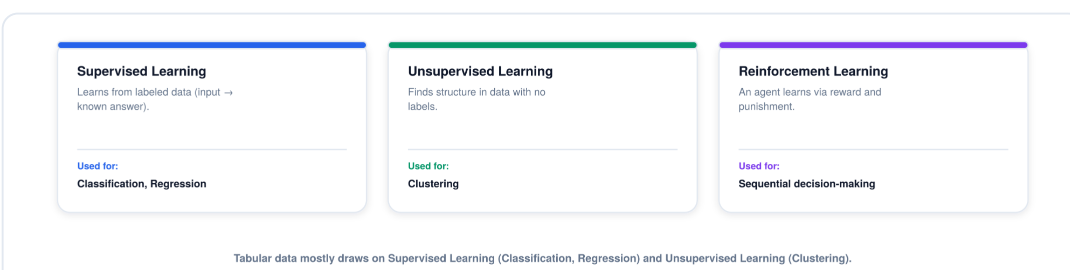


Fig. 14 — The three branches of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

Within these branches, tabular data commonly maps to three specific problems:

- **Classification** (Supervised Learning) — output is binary or multi-class. *e.g. will the customer churn — Yes/No; which membership tier — Bronze/Silver/Gold/Diamond?*
- **Regression** (Supervised Learning) — output is a continuous number. *e.g. predicted house price, predicted monthly revenue.*
- **Clustering** (Unsupervised Learning) — groups unlabeled data by similarity. *e.g. finding customer segments with similar purchasing behavior.*

Tabular data is mainly used for three directions: Classification, Regression, and Clustering.

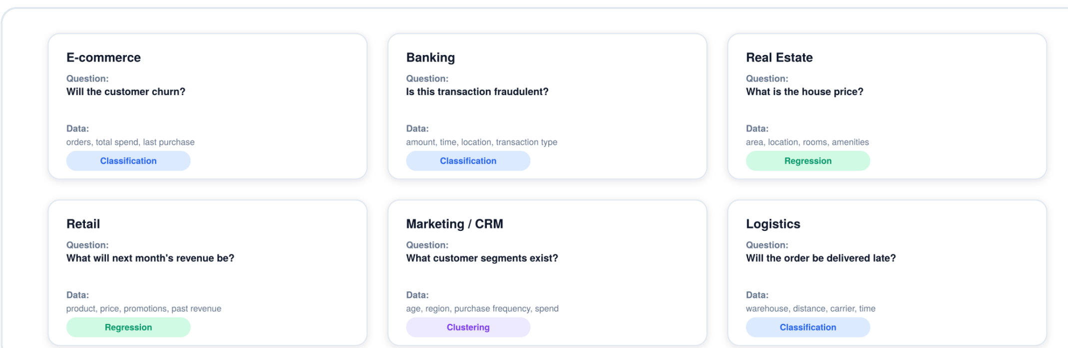


Fig. 15 — Six real-world tabular ML use cases across e-commerce, banking, real estate, retail, marketing, and logistics.

KEY WORDS

classification · regression · clustering

8. Demo with Google Colab

The dataset is created directly inside the notebook, then **pandas** reads this CSV file as a `DataFrame` to practice basic commands. Pandas represents data as a DataFrame — much like an Excel sheet inside Python.

```
file_path = "orders_demo_beginner.csv"
Path(file_path).write_text(csv_content, encoding="utf-8")

df = pd.read_csv(file_path)
```

Core pandas commands from the Colab demo

```
df = pd.read_csv(file_path)
```

Reads the CSV file into a pandas DataFrame.

```
df.shape
```

Returns the table's size as (rows, columns).

```
df.columns
```

Returns the list of column names.

```
df.isna()
```

Returns a True/False mask marking which cells are missing.

```
df.isna().sum()
```

Counts missing values per column — 0 means that column has no missing data.

Example output of df.isna().sum()

Column	Missing count
CustID	0
Age	1
City	1
Orders	0
Spent	0

← non-zero values flag exactly which columns need cleaning

pandas represents data as a DataFrame — much like an Excel sheet inside Python.

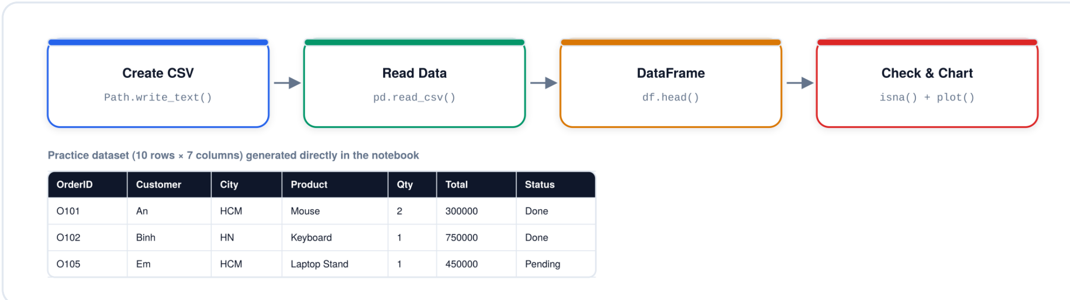
Fig. 16 — Core pandas commands from the demo, and an example df.isna().sum() output.

Code to remember from the demo:

- `df.shape` — tells you the size of the table (rows, columns).
- `df.columns` — tells you the list of column names in the table.
- `df.isna()` — checks which cells are missing data.
- `.sum()` — counts the number of missing values in each column.

If a column's result is 0, it means that column has no missing data:

```
df.isna().sum()
```



OrderID	Customer	City	Product	Qty	Total	Status
O101	An	HCM	Mouse	2	300000	Done
O102	Binh	HN	Keyboard	1	750000	Done
O105	Em	HCM	Laptop Stand	1	450000	Pending

Fig. 17 — The demo pipeline: create a CSV file, read it with pandas, inspect the DataFrame, then check and chart it.

Part III

Time-Series Data

1. Time-Series definition

Time-series data is a set of data points recorded in chronological order.

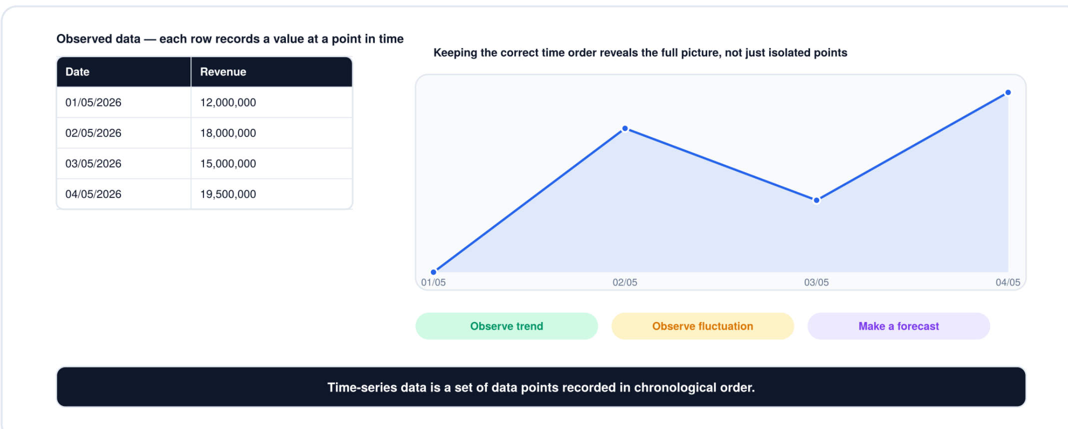


Fig. 18 — Daily revenue table and line chart illustrating that time order reveals trend, fluctuation, and forecasting potential.

2. Timestamps

A timestamp lets us do four things with a series: **establish sequence, measure distance, aggregate data, and identify fluctuation.**

> **Sequencing matters:** data must strictly follow chronological order (day 1, then day 2). Reordering it breaks the underlying meaning and makes forecasting impossible.

KEY WORDS

sequencing · measuring distance · aggregating data · identifying fluctuation

3. Time step, frequency, and granularity

Timestamps & Frequency: data is collected at fixed intervals apart (e.g. taking a stock price at exactly 8:00, 10:00, 12:00 every day).

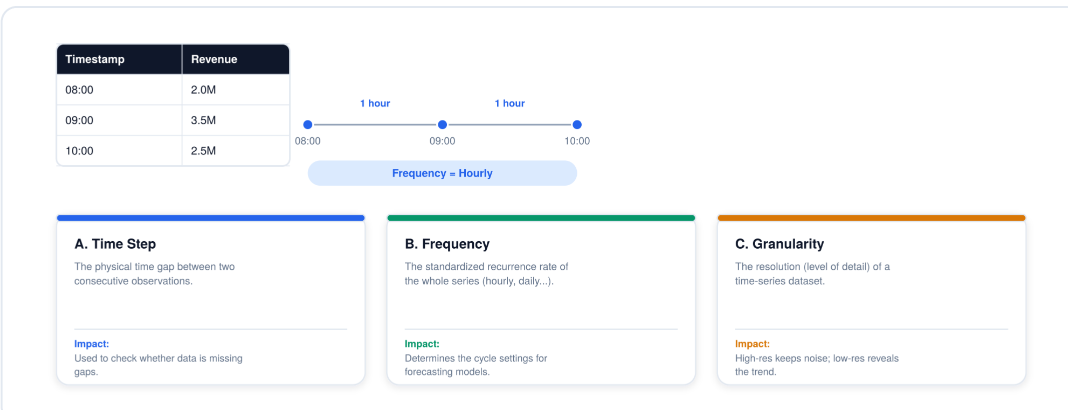


Fig. 19 — Time Step, Frequency, and Granularity explained with an hourly revenue example.

4. Types of Time-Series Data

- **Univariate** — 1 object + 1 variable. *e.g. observing daily profit alone.*
- **Multivariate** — 1 object + multiple variables. *e.g. observing both profit and order count by day to find a correlation.*
- **Multiple Time-Series** — multiple objects + 1 variable. *e.g. tracking Store A's and Store B's revenue over time to compare which grows faster.*

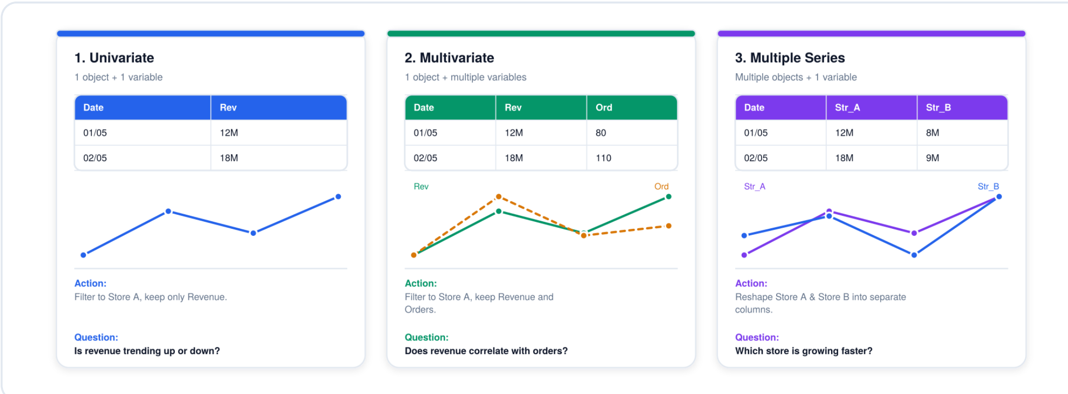


Fig. 20 — Univariate, Multivariate, and Multiple Time-Series compared with mini charts and example tables.

KEY WORDS

univariate · multivariate · multiple time-series

5. Where is time-series data?

Time-series data always serves three big problems: Forecasting, finding a Trend, or detecting an Anomaly.

Point in time → measured value → a trend / forecasting problem

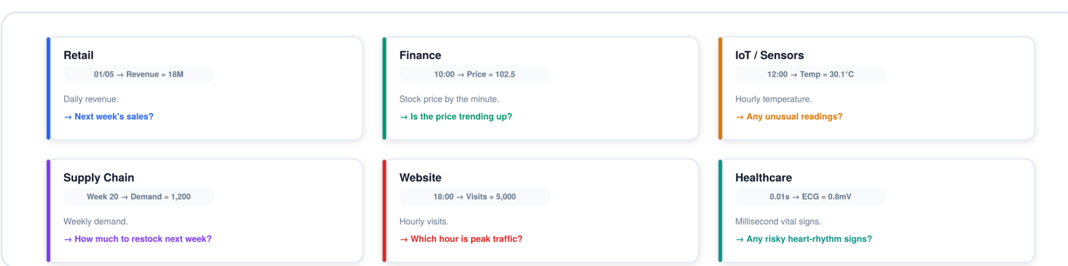


Fig. 21 — Six real-world time-series domains: retail, finance, IoT sensors, supply chain, websites, and healthcare.

a. Forecasting

Step 1 — collect historical data. **Step 2** — learn the pattern in it (trend, seasonality, noise). **Step 3** — forecast the future.

Uses: inventory management, sales planning, staff allocation, budget planning.

Fig. 22 — Forecasting pipeline: collect past data, learn the pattern, forecast the future.

b. Trend

Upward trend vs. **Stationary / no trend.**

Fig. 23 — Upward trend versus a stationary (no-trend) series, side by side.

c. Anomaly

Step 1 — learn the normal value range. **Step 2** — compare each new point to that learned pattern. **Step 3** — flag the anomaly.

Uses: banks (unusual transactions), IoT/sensors (equipment-malfunction warnings), websites/apps (detecting abnormal traffic spikes/drops), warehouses (flagging abnormal stock mismatches).

Fig. 24 — Anomaly detection pipeline: learn the normal range, compare new points, flag anomalies.

6. Trend, Seasonality & Cyclic pattern

- **Trend** — Upward Trend (sloping up) / Stationary (moving sideways, stable).
- **Seasonality** — repeats on a *fixed* cycle (you know exactly when it repeats again). *e.g. Grab ride-hailing demand always peaks at 7–8am and 4–5pm every day.*
- **Cyclic Pattern** — a long-term up-and-down movement through phases (Rise → Peak → Fall → Trough), where the length of time is *not fixed / uncertain*. *e.g. Bitcoin price, stock prices.*

Fig. 25 — Weekly retail revenue with recurring weekend spikes across two full cycles.

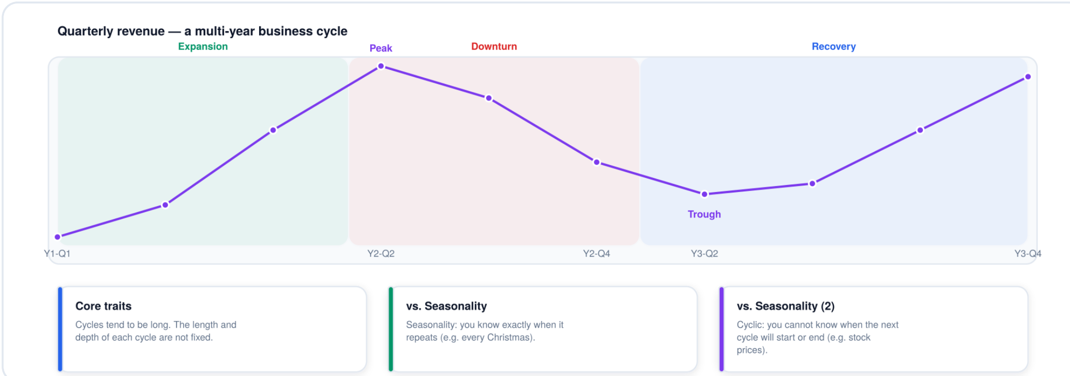


Fig. 26 — A multi-year quarterly revenue cycle moving through expansion, peak, downturn, and recovery phases.

KEY WORDS

trend · seasonality · cyclic pattern (business cycle)

7. Noise vs. Anomaly in Time-Series Data

- **Noise** — small fluctuation around a familiar average, caused by normal random factors in real-world data. Doesn't break the trend, seasonality, or main pattern.
- **Anomaly** — a point that deviates sharply from its neighbors. Could be a data/system error, a stockout, or a shock event (e.g. a website going down).

Fig. 27 — Noise (small random fluctuation) compared side by side with an anomaly (a sharp, isolated deviation).

KEY WORDS

noise · anomaly

8. Related problems: Forecasting & Anomaly Detection

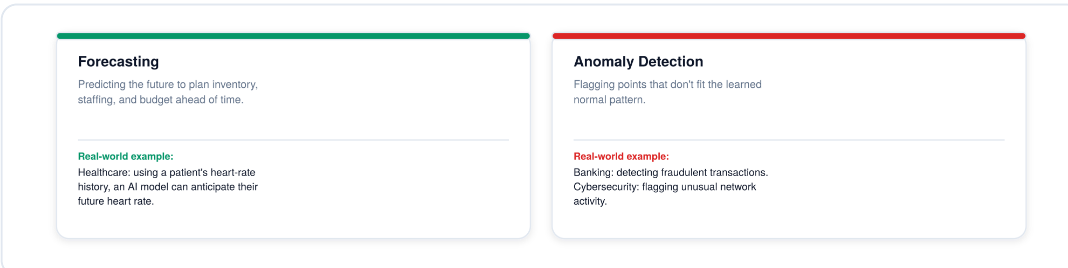


Fig. 28 — Forecasting and Anomaly Detection, with real-world examples for each.

Forecasting — predicting the future ahead of time to manage inventory and staffing. *e.g. in healthcare, based on a patient's heart-rate history, an AI model can anticipate their future heart rate.*

Anomaly Detection — detecting fraudulent transactions, cybersecurity alerts, and more.

See the coding demo in the companion Google Colab file.

Part IV

Summary and Demo

Tabular vs. Time-Series, side by side

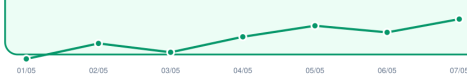
Tabular Data

- Each row is one object or one data sample.
- Each column represents an attribute: age, income, product type.
- Rows are usually relatively independent of each other.
- Row order usually doesn't change the meaning much.
- Good fit for classification, regression, and clustering.

Customer ID	Age	Income	Purchased
KH001	25	15,000,000	Yes
KH002	34	22,000,000	No

Time-Series Data

- Each observation is tied to a specific point in time.
- Chronological order matters a lot and should not be shuffled.
- The current value can be influenced by past values.
- Common in revenue, temperature, sensors, stock prices.
- Good fit for forecasting and anomaly detection.



Tabular focuses on rows & columns; Time-series focuses on chronological order and change over time.

Fig. 29 — Tabular Data and Time-Series Data compared side by side, each with a mini example.

Criterion	Tabular Data	Time-series Data
Organization	Arranged in rows and columns.	Arranged in chronological order.
Observation unit	Each row is one data sample.	Each row is one point in time.
Order matters?	Row order usually isn't critical.	Time order is critical.
Record relationship	Samples are relatively independent.	Current value can depend on past values.
Key columns	Features, label, numeric / categorical cols.	Timestamp, measured value, time variable.
Examples	Customers, products, orders, grades.	Revenue, temperature, stock price.
Goal	Classification, regression, clustering.	Forecasting, anomaly & trend detection.

Tabular example

SKU	Price	Category
SP001	120,000	Electronics
SP002	85,000	Household

Time-series example

Date	01/05	02/05	03/05
Revenue	35	42	38

Conclusion: Tabular focuses on rows & columns; Time-series focuses on time order and change over time.

Fig. 30 — A seven-criterion comparison table between Tabular Data and Time-series Data.

Tabular data cares about rows and columns; time-series data cares about chronological order and how values change over time.

Chart cheat-sheet

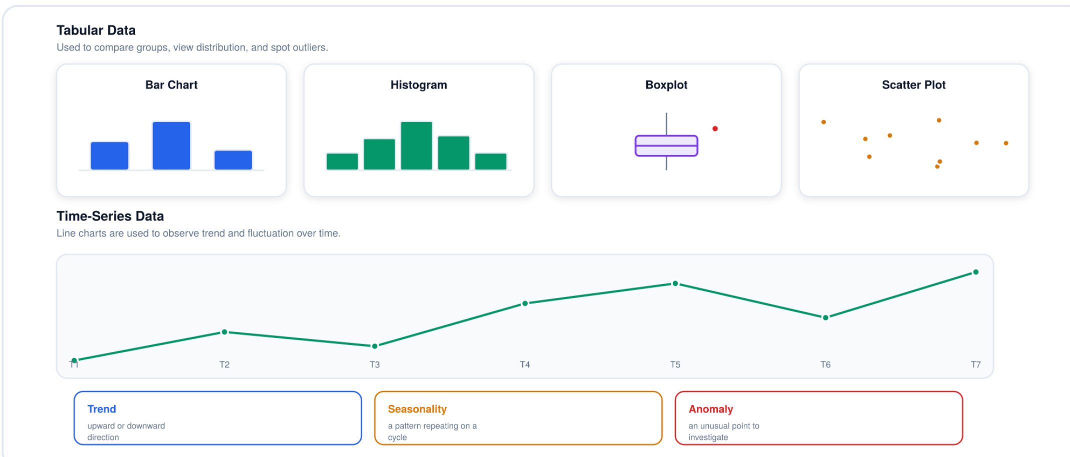


Fig. 31 — Chart-type summary: Bar Chart, Histogram, Boxplot, Scatter Plot for tabular data; Line Chart with Trend, Seasonality, Anomaly for time-series data.

Quick picks: comparing groups → Bar Chart. Viewing a distribution → Histogram. Data over time → Line Chart. Finding a strange point → Boxplot / anomaly detection.

From data to the right ML problem

The first and most important question is always: **what does the output need to predict?**

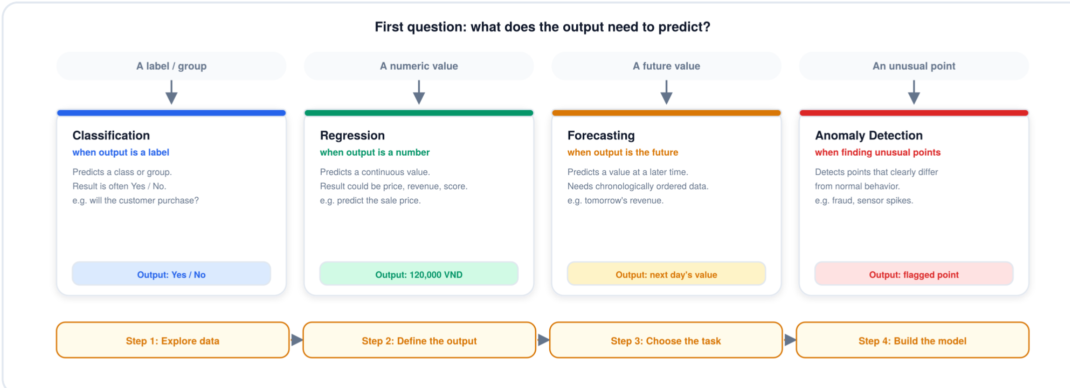


Fig. 32 — Decision flow from output type to Classification, Regression, Forecasting, or Anomaly Detection, plus a four-step thinking process.

- **A label / group → Classification** — e.g. will the customer purchase? Output: Yes/No.
- **A numeric value → Regression** — e.g. predicted sale price. Output: 120,000 VND.
- **A future value → Forecasting** — needs chronologically ordered data. e.g. tomorrow's revenue.
- **An unusual point → Anomaly Detection** — e.g. fraud, a sensor spike.

General thinking process: **explore the data → define what the output should be → choose the matching ML task → build and evaluate the model.**

These notes summarize the "Tabular & Time-series Data" session from the AI VIETNAM course (M01W01), compiled for personal review. All diagrams were redrawn for this document.